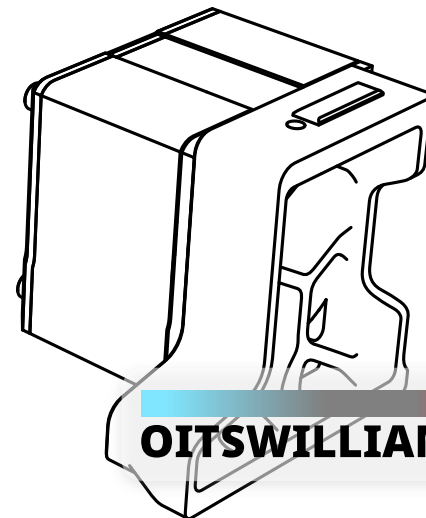
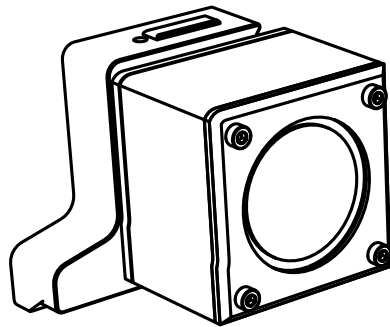


OITSWILLIAMV2 Banger speaker skirts

Assembly guide

REV 2026-01-08



OITSWILLIAM PANG

Hardware

Contents

• <u>Before getting started</u>	3
• <u>Speaker wiring</u>	5
• <u>Assembly</u>	6
• <u>Exploded view</u>	
• <u>Detailed assembly</u>	7

Before getting started

You are about to read the manual of banger speaker skirts. This manual details about it and assembly of such speakers.

About the banger speakers

This mod adds speakers that sound as loud as it gets, with 30mm radiators to improve bass. It is mounted to Voron 2 150mm (mod). However users can access the CAD model of such model to attach them to other units.

Contribution

This mod is a part of OITSWILLIAMV2 mod project through GitHub. You can visit, contribute and report new issues here.

Technical support

You can contact Oitswilliam here, or if the issue is not related to OITSWILLIAMV2 mod, you may visit Voron forum or its Discord server.

Warning

-  Danger: unless if it only runs on DC and only has a DC input, otherwise risk of electric shock
- Even though you can build with this mod from start, if you want to wire, or re-wire, you need to make sure and check that the mains is unplugged.
 - Failure doing so which is unplugging the mains power may cause electrocution or even death.
 - When in doubt, please don't touch the wires connected to the power supply or SSR.

Other notes

- **There are two parts of per model on each side of the skirts**, these being the main part and the modifier. The modifier is used for adding the mesh on the part where it has ventilation or speaker cutouts. Please visit the parts' Readme file on how to prepare these.

- The righthand side of the speaker has another notch that is located at the top, whereas the lefthand side doesn't. This indicates the side and is used for alignment. If the righthand side parts are attached to the lefthand sides, or vice versa, it will be an obvious telltale sign that is incorrect. The box units contain two letters that indicate their sides.
- For their abbreviations, LH means lefthand side and RH means righthand side.

Guide and legend

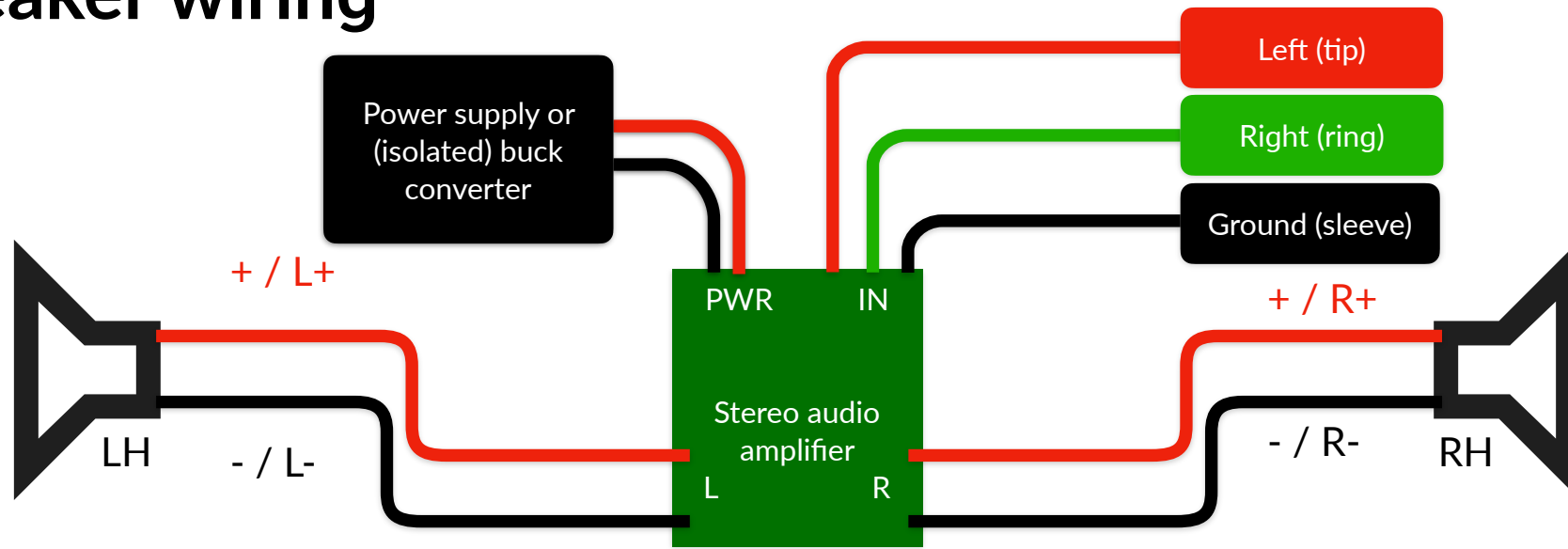
-  - raw material
-  - highlighted material
-  - printed material, as this part is simple, the printed parts may be color-coded.

Part printing

The parts of this model can be printed in the base color of the printer. **Bold text** indicates the recommended value and *italic text* indicates on what setting / value it needs to set.

FDM / FFF or SLS	Print material: ABS, ASA or PA12, prohibits PLA.
Layer height of 0.2mm , initial layer height can be 0.2mm or higher.	<i>Nozzle and extrusion width</i> must be forced 0.4mm; if using 0.4mm nozzle
1.6mm wall width , or 4 wall counts.	<i>Infill</i> must be at least 40%, except on the infill modifiers.
At least 4 solid bottom and top layers, except on the infill modifiers.	The best <i>print speed</i> is up to 120mm/s . The speed might be reduced by the hot end flow rate.
<i>Supports</i> are recommended for main skirt parts. Use tree or organic supports if available.	<i>Other settings might be depended on your printer.</i> <i>Follow it's warnings for proper handling.</i>

Speaker wiring

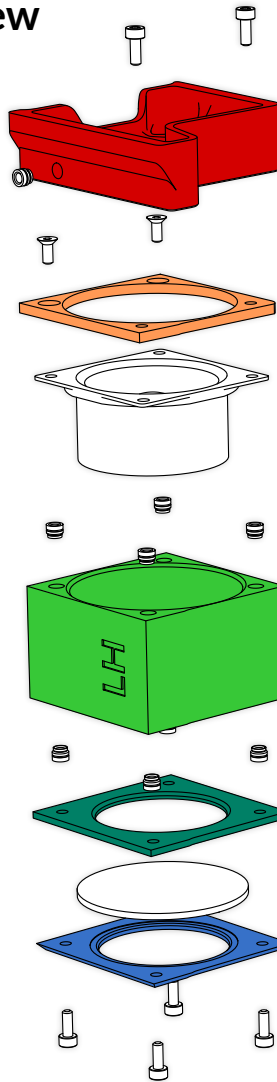


- Note that the speaker wire polarity must be matched in order to function correctly, and should not be reversed.
- The power requirements for the amplifiers are different, some of them accept 5V, while others accept 12V or 24V.
- **If the speakers did not have tails of wires, you may make them yourself.** The wire length should be at least 250mm per connection, totaling 1000mm. (On bigger machines the length may be added more)
- **The wiring colors for the analog TRS plug may be different,** because of this, you may use a multimeter to check whether the wire and the jack match.
 - For left (tip), it may use red or blue, for right (ring), it may use green, red or white and for grounding (sleeve) it may be either exposed, or covered with green or black insulation.
- After done wiring and testing, you may unplug them, and prepare them for installation to the printer.

Assembly

Exploded view

M3*5*4 heat set inserts



Lefthand

Per sides

2* M3*8mm SHCS

Side mount skirt

side_skirt_150_speaker_left.stl
side_skirt_150_speaker_right.stl

2* M3*6mm FHCS countersunk

Speaker bracket

left_speaker_unit_bracket.stl
right_speaker_unit_bracket.stl

2-inch main speaker unit

4* M3*5*4 heat set inserts

Speaker unit box

left_speaker_unit_box.stl
right_speaker_unit_box.stl

Wire pass-through

4* M3*5*4 heat set inserts

Speaker radiator inner

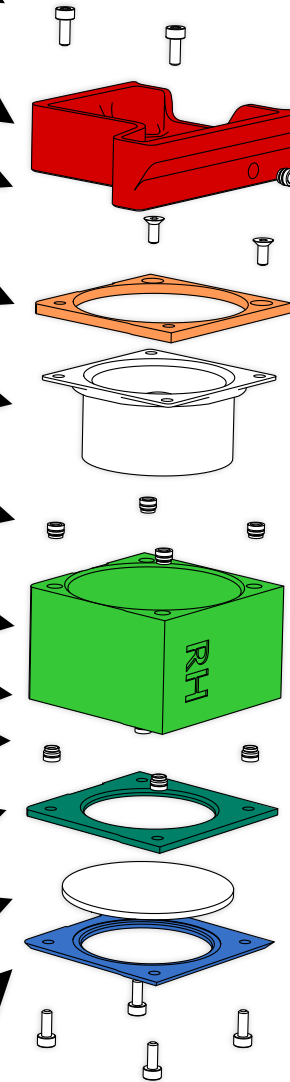
left_speaker_unit_radiator_bracket.stl
right_speaker_unit_radiator_bracket.stl

30mm speaker radiator

Speaker radiator mount

left_speaker_unit_radiator_bracket_mount.stl
right_speaker_unit_radiator_bracket_mount.stl

4* M3*8mm SHCS
or BHCS



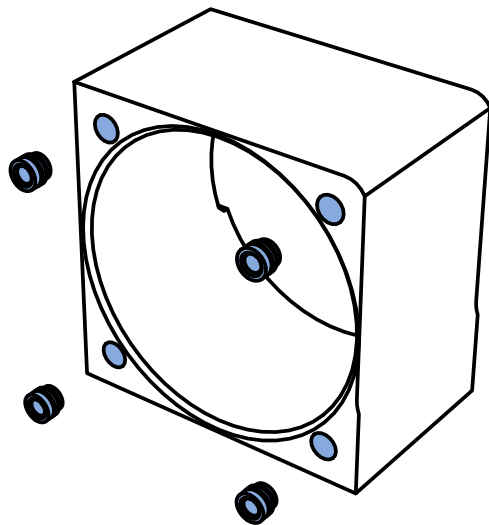
M3*5*4 heat set inserts

Righthand
(With a top notch)

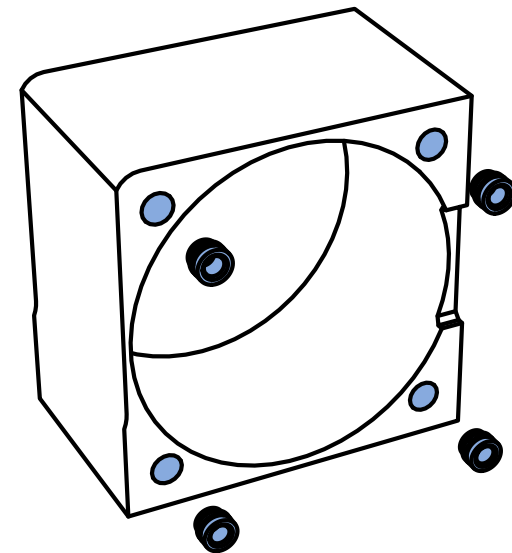
Detailed assembly

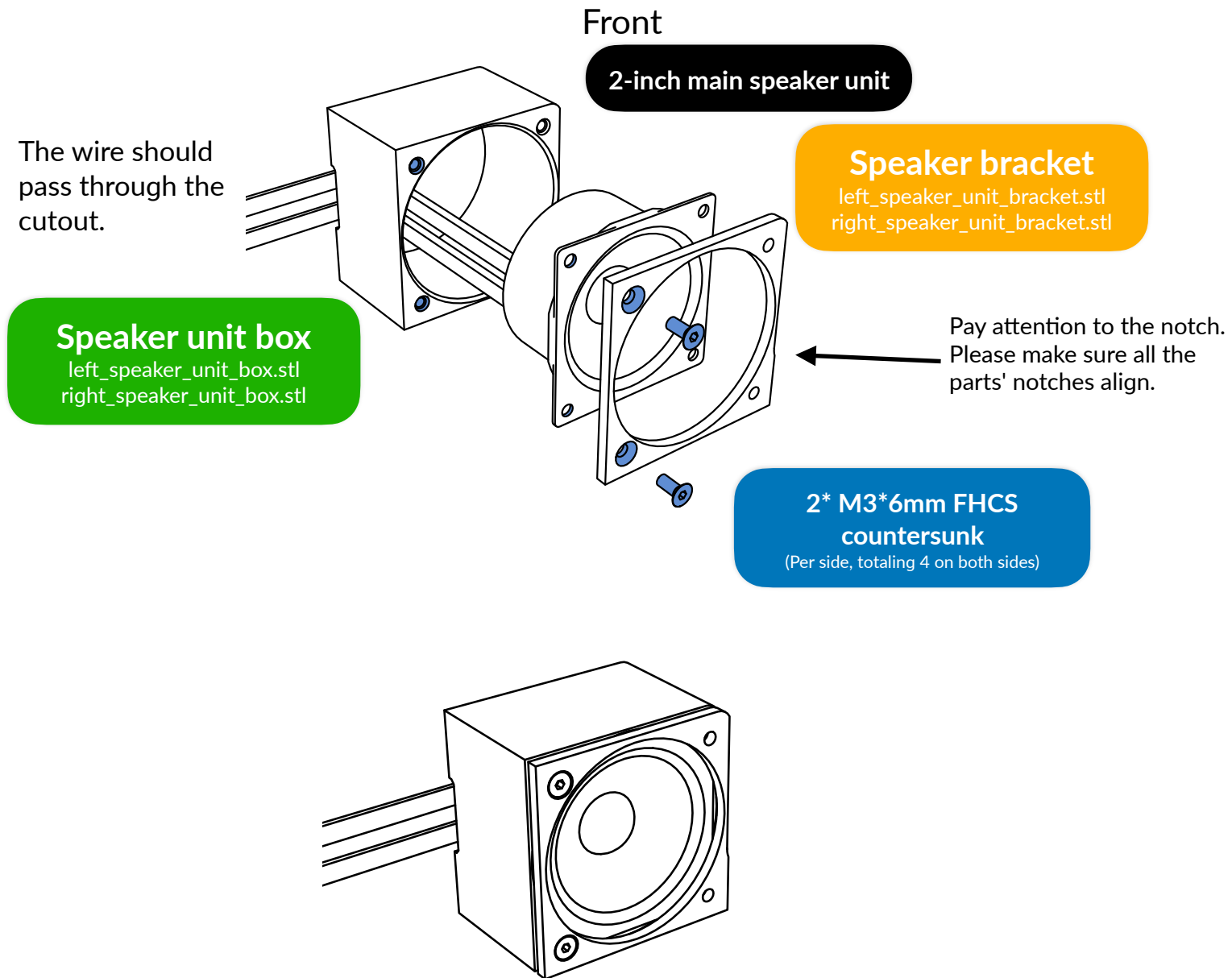
Speaker unit

Showing lefthand side only as righthand side is just mirrored and will follow the same pattern, albeit the righthand side has another notch.



8* M3*5*4 heat set inserts
(Per side, totaling 16 on both sides)





Speaker radiator

Pay attention to the notch.
Please make sure all the
parts' notches align.

Speaker radiator inner

left_speaker_unit_radiator_bracket.stl
right_speaker_unit_radiator_bracket.stl

30mm speaker radiator

Speaker radiator mount

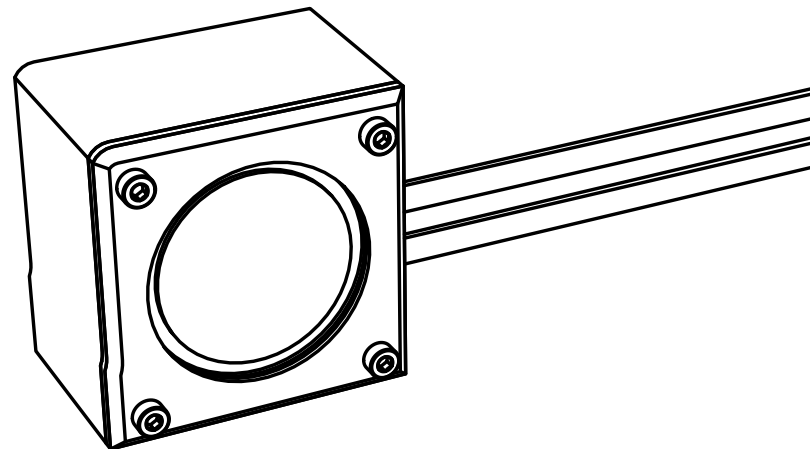
left_speaker_unit_radiator_bracket_mount.stl
right_speaker_unit_radiator_bracket_mount.stl

Back

The wire should
pass through the
cutout.

4* M3*8mm SHCS or BHCS

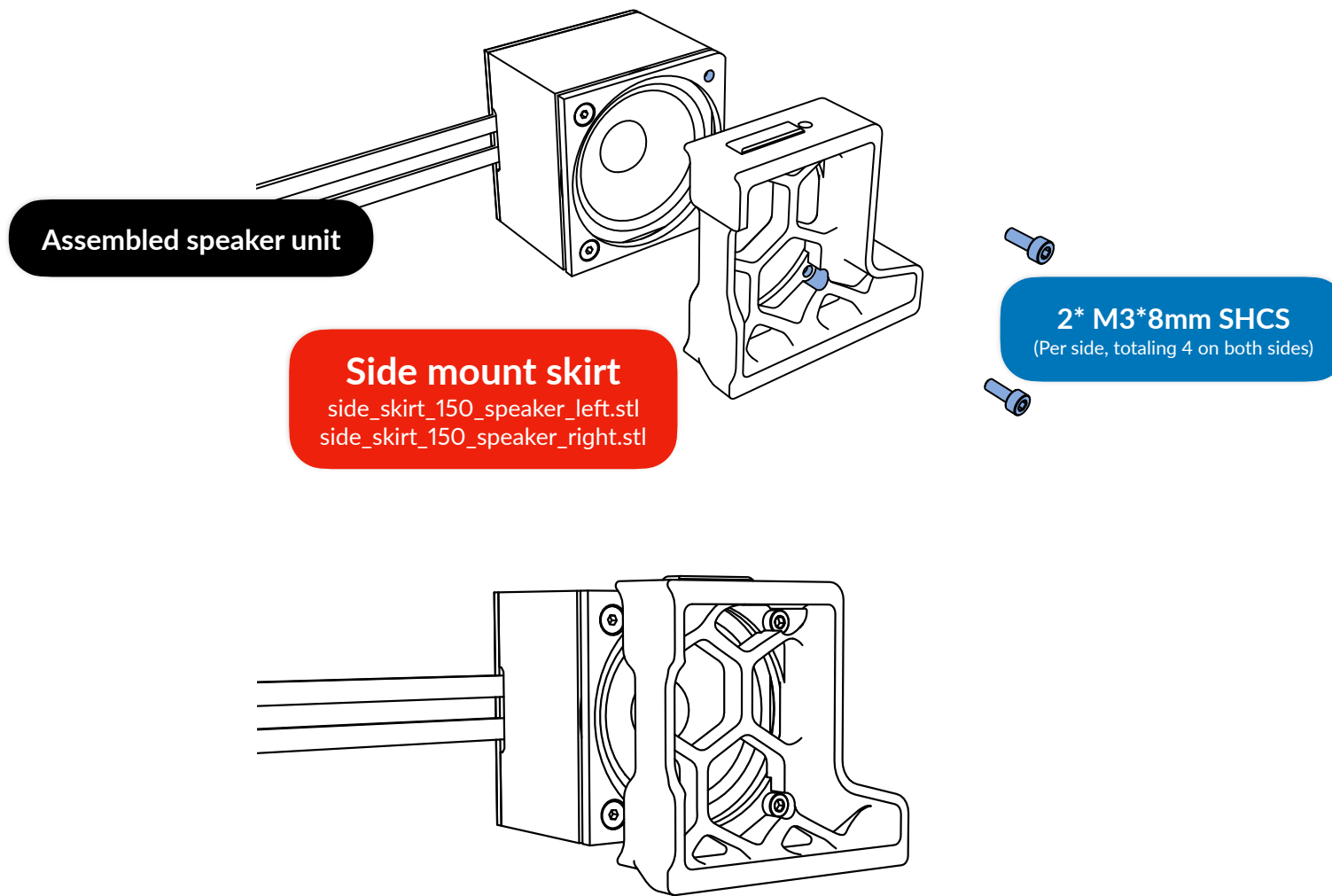
(Per side, totaling 8 on both sides)



The speaker unit
has assembled.

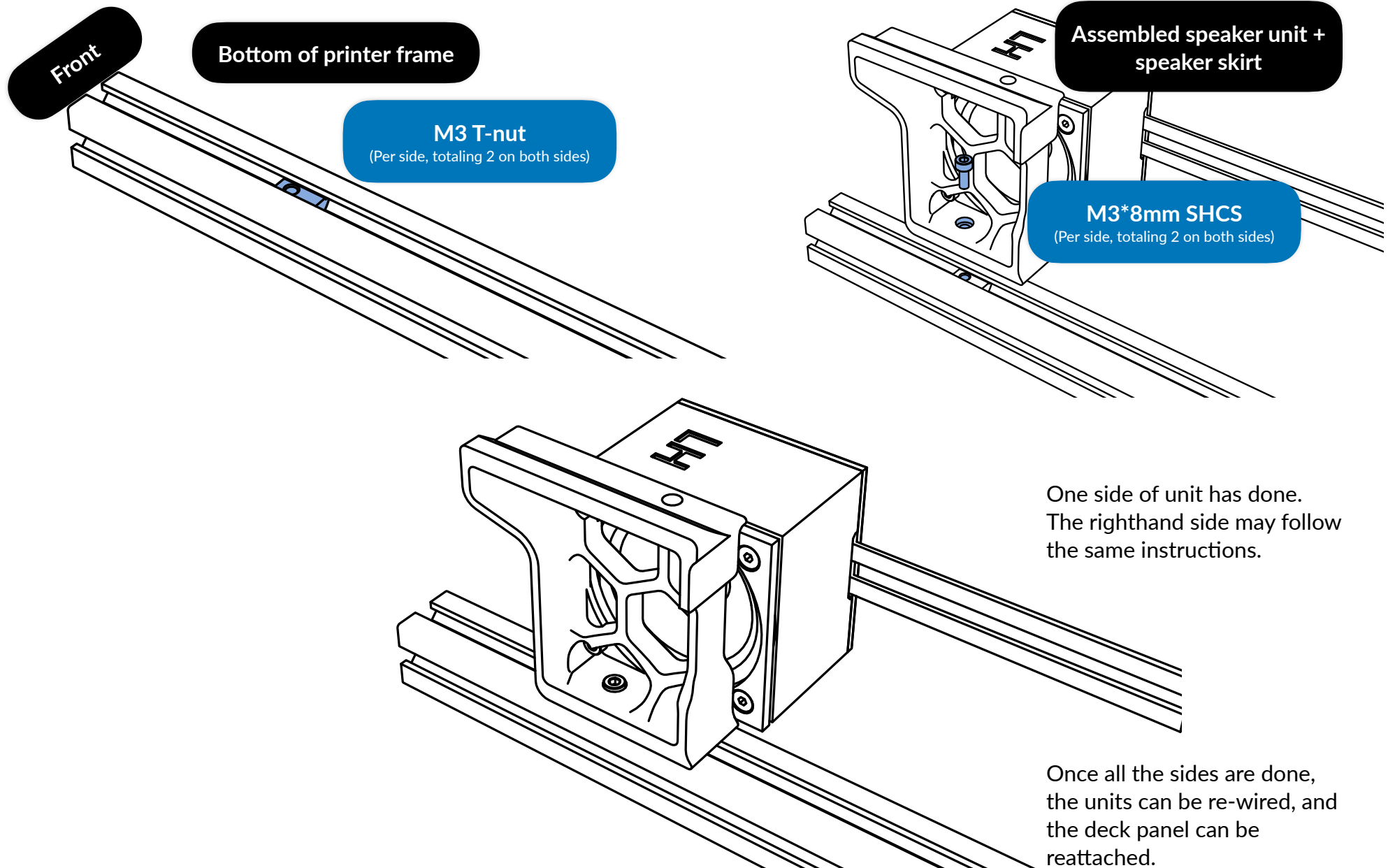
Speaker skirt

These skirts can be followed by page 226 and 227 of the original V2 assembly manual, with just a quick summary, these skirts can be mounted via M3*8mm screws.



Mounting to frame

Based of page 226 and 227 of original V2 assembly manual.



Assembly complete

And we are done! If you are satisfied with such product consider tagging @officialbunny350 on Facebook or @realBunny350 on X (formerly known as Twitter) and share such product with us on Oitswilliam Pang Discord Server.

If you have enquires, problems or issues related to building or using such product, please report it on [GitHub.com/Bunny350/OITSWILLIAMV2/issues](https://github.com/Bunny350/OITSWILLIAMV2/issues) and we will get on it.

Change log

- 2026-01-08 - initial release

Manual designed by Oitswilliam Pang.
One-person operated design.
www.oitswilliam.com